

CYLAB SECURITY ACADEMY

Challenge Write-Up Template

Challenge Information

Student Name	Evallo, Daniella Kim N.
Section / Class	BSIT-3G
Challenge Name	SUDO MAKE ME A SANDWICH
Category	General Skills
Points / Difficulty	Easy
Date Solved	08/10/2026
Flag Format Given	picoCTF{...}

1. Challenge Description

The challenge asks me to connect to a remote Linux environment through SSH and find a way to read the protected flag.txt file. The initial attempt to read the file is denied, so the goal is to inspect the user's sudo permissions and use an allowed program to access the file.

2. Initial Analysis / Reconnaissance

After connecting to the challenge instance, I listed the files in the home directory and found flag.txt. Trying to read it directly with cat flag.txt returned Permission denied. I then checked my sudo permissions with sudo -l. The output showed that ctf-player could run /bin/emacs with sudo without a password, which provided a path to open the protected flag.txt file.

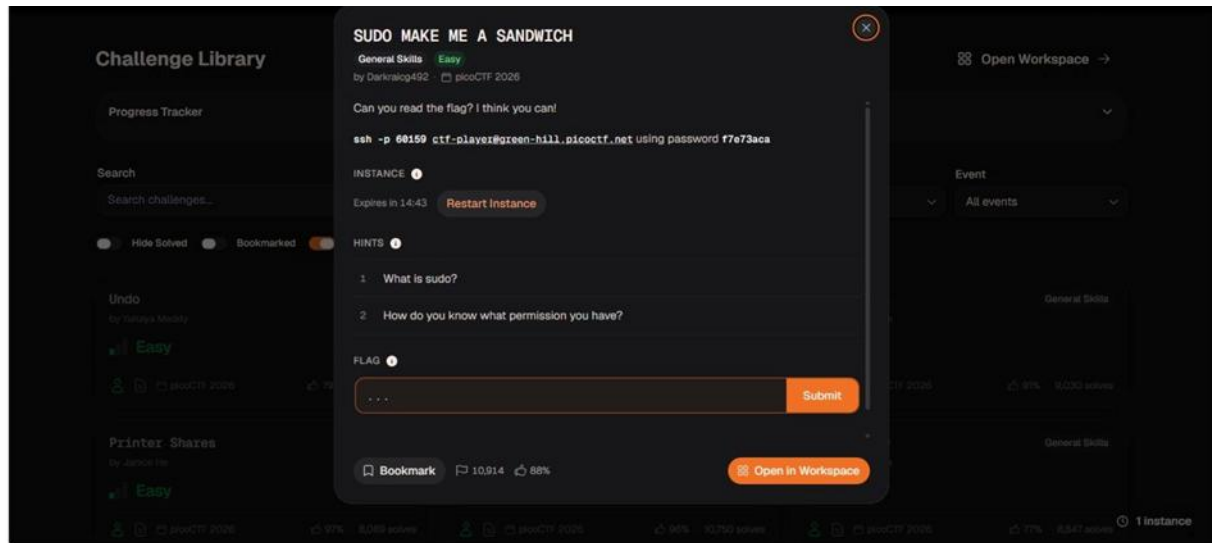
3. Tools and Commands Used

Tool / Command	What it was used for	Result / Output
ssh -p 60159 ctf-player@green-hill.picoctf.net	Connect to the remote challenge environment.	The ctf-player shell was opened.
ls -la	Inspect the files and permissions in the home directory.	flag.txt was visible and owned by root.
cat flag.txt	Attempt to read the flag directly.	Permission denied.
sudo -l	Check which commands the ctf-player account may run with sudo.	NOPASSWD: /bin/emacs was allowed.

4. Solution Walkthrough

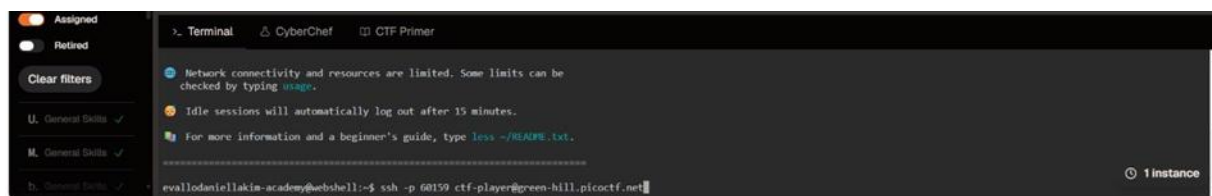
Step 1

I first reviewed the SUDO MAKE ME A SANDWICH challenge page. The prompt provides the SSH connection details, while the hints point toward sudo and checking the permissions available to the current user.



Step 2

From the terminal, used the SSH command provided by the challenge: `ssh -p 60159 ctf-player@green-hill.picoctf.net`. The challenge also provides the password `f7e73aca` for the ctf-player account.



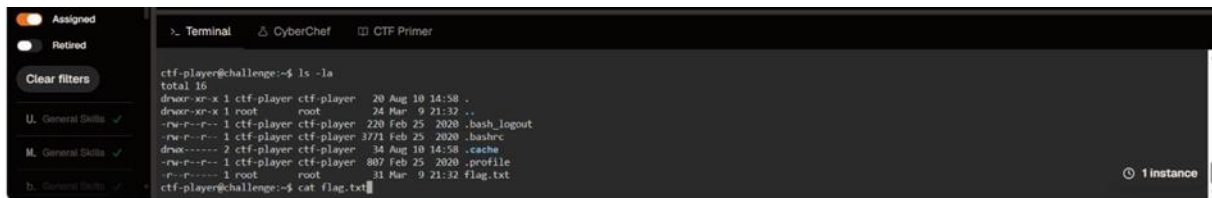
Step 3

Open the active workspace and confirmed that the challenge instance was running. The terminal was ready for the SSH connection and the SSH command `ssh -p 60159 ctf-player@greenhill.picoctf.net`. The `-p 60159` option specifies the challenge port, while the username and hostname identify the remote machine.



Step 4

After logging in, ran `ls -la` to inspect the home directory. The listing showed `flag.txt` and its permissions. then attempted to read it using `cat flag.txt`.



```
ctf-player@challenge:~$ ls -la
total 16
drwxr-xr-x 1 ctf-player ctf-player 20 Aug 10 14:58 .
drwxr-xr-x 1 root      root      24 Mar  9 21:32 ..
-rw-r--r-- 1 ctf-player ctf-player 220 Feb 25 2020 .bash_logout
-rw-r--r-- 1 ctf-player ctf-player 3771 Feb 25 2020 .bashrc
drwx----- 2 ctf-player ctf-player 34 Aug 10 14:58 .cache
-rw-r--r-- 1 ctf-player ctf-player 807 Feb 25 2020 .profile
-rw-r--r-- 1 root      root      31 Mar  9 21:32 flag.txt
ctf-player@challenge:~$ cat flag.txt
```

Step 5

The direct `cat flag.txt` command returned `Permission denied`. then used `sudo -l` to determine which commands the `ctf-player` account was allowed to run with elevated privileges.



```
ctf-player@challenge:~$ cat flag.txt
cat: flag.txt: Permission denied
ctf-player@challenge:~$ sudo -l
Matching Defaults entries for ctf-player on challenge:
  env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User ctf-player may run the following commands on challenge:
  (ALL) NOPASSWD: /bin/emacs
ctf-player@challenge:~$
```

Step 6

The `sudo -l` output showed that the account could run `/bin/emacs` without entering a password. I then used the permitted command `sudo /bin/emacs -nw flag.txt` to open the protected file with elevated privileges.



```
ctf-player@challenge:~$ cat flag.txt
cat: flag.txt: Permission denied
ctf-player@challenge:~$ sudo -l
Matching Defaults entries for ctf-player on challenge:
  env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User ctf-player may run the following commands on challenge:
  (ALL) NOPASSWD: /bin/emacs
ctf-player@challenge:~$ sudo /bin/emacs -nw flag.txt
```

Step 7

The protected `flag.txt` file opened in Emacs, and the picoCTF flag was visible in the editor.



```
File Edit Options Buffers Tools Text Help
picoCTF{ju57_5ud0_37_94183884}

-- 11 flag.txt All 11 (Text)
For information about GNU Emacs and the GNU system, type C-h C-a.
```



5. Flag

```
picoCTF{ju57_5ud0_17_9418380d}
```

6. Key Concept / What I Learned

I learned how sudo permissions can be used to access a protected file when a user is allowed to run a specific program with elevated privileges. In this challenge, flag.txt could not be read directly because of its permissions, but sudo -l revealed that /bin/emacs could be executed without a password. Running Emacs with sudo allowed the protected flag.txt file to be opened and the flag to be recovered.

7. References

[picoCTF 2026 | Denton's Blog](#)

YouTube – picoCTF 2026 / CyLab – Sudo make me a sandwich

<https://youtu.be/dJwoUj9p--E?si=Dh1BaQpyaEuaa61r>